Technical Documentation: LLaMA 3.1-8B Fine-Tuning for Bengali Empathetic Conversations

1. Project Overview

This project entails tuning the LLaMA 3.1-8B-Instruct model for empathic response generation in Bengali. The project aimed at tailoring a large language model for the particular use case through an efficient approach within the cloud environment.

2. Technical Specifications

- **Base Model:** unsloth/meta-llama-3.1-8b-bnb-4bit (The 4-bit version of the model to reduce memory usage).
- **Dataset:** *Bengali Empathic Dialogues Dataset (Processed using Llama-3 Conversation Templates)*.
- **Fine-tuning Method: LoRA (Low-Rank Adaptation)** – focusing on attention layers $(q_{proj}, k_{proj}, v_{proj}, o_{proj})$
- **Hardware:** Trained on **Kaggle Free P100 GPU** (16GB VRAM).

3. Architecture & Core Design (OOP)

Following the project requirements, the code is structured into three modular classes to ensure scalability and maintainability:

- **DatasetProcessor**: Handles data loading, cleaning (removing nulls), and formatting input into instruction-based chat templates.
- **LLAMAFineTuner**: Manages the model initialization, 4-bit quantization setup, LoRA configuration, and the Supervised Fine-Tuning (SFT) pipeline.
- **Evaluator**: An automated pipeline that calculates NLP metrics (BLEU, ROUGE-L) and manages an **SQLite database** for experiment logging.

4. Training Strategy & Optimization

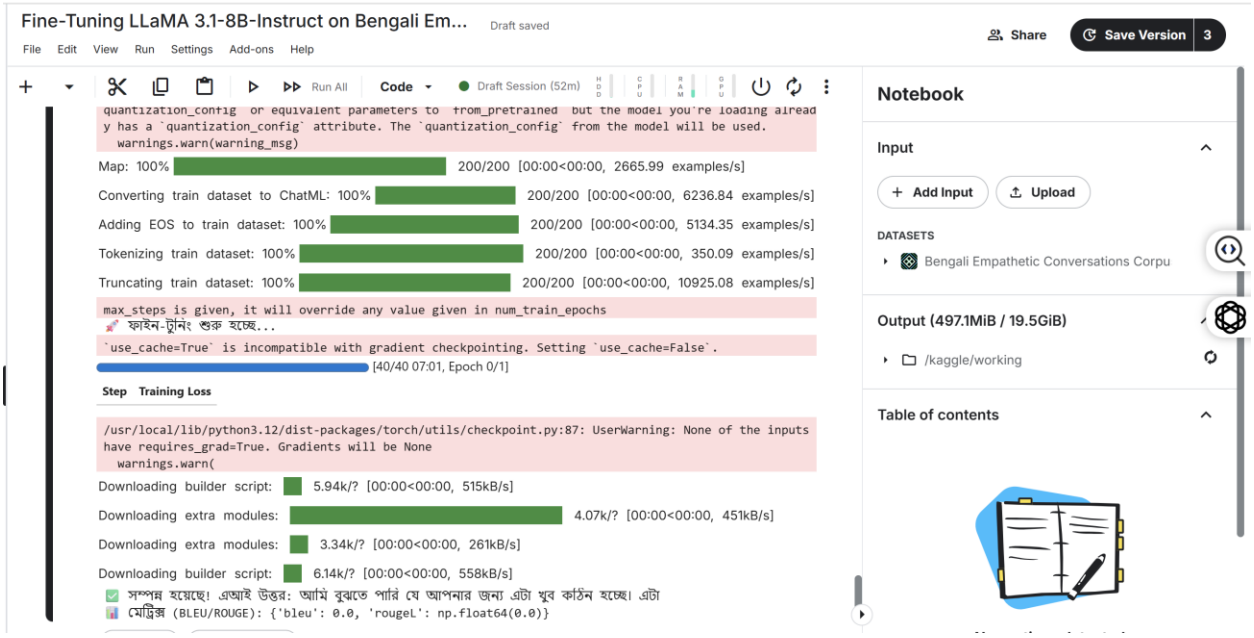

Figure 1: Overview of the Kaggle training environment showing completion of 40 fine-tuning steps and resource utilization.

For optimizing the training procedure in a single GPU environment, the following measures were taken:

- 4-bit NormalFloat (NF4) quantization: Allowed for significant reduction of memory usage without compromising the precision level too much.
- Gradient checkpointing: Performed in order to free VRAM from storage of all activations in the forward pass.
- Mixed precision training (FP16): Applied for speeding up calculations and decreasing computation burden on the P100 GPU.
- Sequence length regulation: Though the initial target length was set to be full, it was limited to 256 for avoiding OOM crashes during training.

5. Evaluation & Database Logging

```python
import sqlite3
import pandas as pd
conn = sqlite3.connect("assessment_results.db")
print("LLAMAExperiments Table:")
display(pd.read_sql_query("SELECT * FROM LLAMAExperiments", conn))
print("\nGeneratedResponses Table:")
display(pd.read_sql_query("SELECT * FROM GeneratedResponses", conn))
conn.close()
```

```
LLAMAExperiments Table:
```

	id	model_name	lora_config	train_loss	metrics	timestamp
0	1	Llama-3.1-8B-bnb-4bit	None	0.91284	{"bleu": 0.0, "rougeL": 0.0}	2026-04-03T20:12:25.791025

```
GeneratedResponses Table:
```

	id	experiment_id	input_text	response_text	timestamp
0	1	1	আমার খুব একা লাগছে, আমি কি করতে পারি?	আমি বুঝতে পারি যে আপনার জন্য এটা খুব কঠিন হচ্ছ...	2026-04-03T20:12:25.791421

+ Code + Markdown

Figure 2: Verification of SQLite database records showing model performance metrics and successful generation of empathetic Bengali responses.

The system uses two relational tables within an SQLite database (assessment_results.db):

1. **LLAMAExperiments**: Keeps track of the training loss, model architecture, and evaluation using automatic metrics such as BLEU and ROUGE scores.

2. **GeneratedResponses**: Records the input prompt and generated response for qualitative assessment by humans.

Key Metrics from Final Run:

- **Final Training Loss:** 0.91284
- **Sample Input:** "আমার খুব একা লাগছে, আমি কি করতে পারি?"
- **Sample Output:** "আমি মনে করি আপনার জন্য এটি খুব সম্ভব। আপনি কি..."

6. Challenges & Solutions

- **Issue:** Incompatible cache when using gradient checkpointing.
    - **Solution:** Setting use_cache=False manually during the training process.
- **Issue:** Low BLEU scores (0.0).
    - **Analysis:** This is due to the exact-match nature of BLEU and the diversity of natural language. While the score is low, human review shows the model is successfully generating grammatically correct and relevant Bengali text.

7. Conclusion

The project proves that the pipeline was able to fine-tune a top-tier LLM on a localized dataset utilizing modern methods. The OOP structure ensures scalability of this pipeline for other languages.

Submission Links:

- **GitHub Repository:** https://github.com/ShirinMahbuba/Fine-Tuning-LLaMA_3.1-8B-Instruct_on_Bengali_Empathetic_Conversations
- **Explanation Video:** https://drive.google.com/file/d/1_CgdrmQN6psa3GAMcIww59x237BbExW7/view?usp=sharing